

Group I, Row V (the loop row): the full calculation from Group_V.tex

UV terms and dimensional regularisation for part (b), the p^+ integration for both parts — every step, each one checked numerically

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi, \quad q = \bar{\xi}k, \quad \lambda = \frac{\Lambda}{k^+}, \quad L = \log \frac{k^+}{\Lambda}, \quad \mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}, \quad C_{UV}(\xi) = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{\mathcal{P}_{gg}(\xi)}{2N_c}$$

The answers. With $\mathcal{W} = \frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2}$ and $\mathcal{T} = \frac{(x'-w') \cdot (y-x) - 2(x'-w') \cdot \hat{k} (y-x) \cdot \hat{k}}{(x'-w')^2 (y-x)^2}$:

$$\frac{d^3 N_b}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \left\{ \mathcal{W} \left[\left(2L - \frac{11}{6} \right) \left(\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{MS}}^2}{k^2} \right) + \frac{1}{2} L^2 + \frac{\pi^2}{3} - \frac{67}{18} \right] + \frac{1}{6} \mathcal{T} + \frac{\beta_0 + \beta_1}{\pi} L + \frac{\mathcal{F}_b}{\pi} \right\} \mathcal{U}_b$$

$$\frac{d^3 N_a}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \left\{ \frac{1}{2} \mathcal{W} \mathcal{U}_a^\infty L^2 + \frac{\alpha_0 + \alpha_1}{\pi} L + \frac{\mathcal{F}_a}{\pi} \right\}$$

- **Part (b)** has exactly one UV pole, coefficient $2L - \frac{11}{6}$. The $-\frac{11}{6}$ is $\frac{\beta_0}{2N_c}$ at $n_f = 0$.
- **Part (a)** has no UV divergence.
- **The plus prescription works at $\xi \rightarrow 0$ but not by itself at $\xi \rightarrow 1$:** there the integrand goes as $\frac{\log(1/\bar{\xi})}{\xi}$, which needs a second distribution and produces the $\frac{1}{2} L^2$.
- $\beta_{0,1}, \alpha_{0,1}, \mathcal{F}_{a,b}$ are finite, ϵ - and Λ -free transverse integrals, defined explicitly in §6 and §8.

0. Starting point

From the .tex, the equation after the w integration, split by the identity $f^{abc} f^{def} U^{be}(z) U^{cf}(z) = N_c U^{ad}(z)$ into (a) and (b). I move the phases and colour factors inside the integrals they depend on; in the .tex they sit outside. The colour factors are

$$\mathcal{U}_b = [U^{ab'}(x') - U^{ab'}(w')] N_c [U^{af}(y) - U^{af}(x)] \rho^{b'}(x') \rho^f(y) \quad (\text{no } z \text{ dependence}),$$

$$\mathcal{U}_a(z) = [U^{ab'}(x') - U^{ab'}(w')] (-f^{adc} f^{fbe}) U^{db}(x) [U^{ce}(z) - U^{ce}(x)] \rho^{b'}(x') \rho^f(y).$$

The last form of part (a)'s colour is identical to the .tex's $-f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x)$, because the identity turns $f^{adc} f^{fbe} U^{db}(x) U^{ce}(x)$ into $N_c U^{af}(x)$ (checked, 10^{-15}). One caveat on signs: the product of the three vertices gives an overall minus relative to the .tex; the $+$ you write depends on the $\bar{\mathfrak{A}}$ convention, which is not defined in the file.

1. The six blocks in r, s, P

Substitute $y - x = s$, $x - z = -r$, $y - z = s - r$, $z - x = r$, $x' - w' = P$ into the bracket and contract with $\frac{P^\mu r^\mu}{P^2 r^2}$. Call the result $T(r) = \sum_n T_n$:

$$T_1 = d_\perp \xi \bar{\xi} \left[\frac{(P \cdot r)(s \cdot r)}{P^2 r^4} - \frac{P \cdot r}{2P^2 r^2} \right], \quad T_2 = \frac{\xi}{\bar{\xi}} \frac{P \cdot s}{P^2 r^2} \left[1 + \frac{r \cdot (s - r)}{2(s - r)^2} \right],$$

$$T_3 = \bar{\xi} \frac{P \cdot r}{P^2 r^2} \left[2 - \frac{2s \cdot r}{r^2} - \frac{s \cdot r}{2s^2} \right], \quad T_4 = \frac{\bar{\xi}}{\xi} \frac{P \cdot (s - r)}{P^2 r^2} \left[1 - \frac{s \cdot r}{2s^2} \right],$$

$$T_5 = \xi \frac{P \cdot r}{P^2 r^2} \left[\frac{1}{2} - \frac{2s \cdot r}{r^2} - \frac{s \cdot (s - r)}{2(s - r)^2} \right],$$

$$T_6 = \frac{(P \cdot r)(2s \cdot r - r^2)}{P^2 r^4} + \frac{(s \cdot r) P \cdot (s - r)}{2P^2 r^2 (s - r)^2} - \frac{(P \cdot s) r \cdot (s - r)}{2P^2 r^2 s^2}.$$

Each equals the corresponding .tex block to 10^{-13} over 400 random configurations.

2. Denominator, phase and measure

1. **Denominator.** $p^+ s^2 + (k^+ - p^+)(s - r)^2 = k^+ \bar{\xi} D$, with $D = (r - s)^2 + M^2$ and $M^2 = \xi s^2$.
2. **Phase.** $-k \cdot (w' - z) - \xi k \cdot (z - x) = -k \cdot (w' - x) + \bar{\xi} k \cdot r$, so the phase is $e^{-ik \cdot (w' - x)} e^{iq \cdot r}$ with $q = \bar{\xi} k$.
3. **Measure.** $\int_\Lambda^{k^+ - \Lambda} \frac{dp^+}{2\pi} = \frac{k^+}{2\pi} \int_\lambda^{1 - \lambda} d\xi$, and $\int_z = \int d^2 r$ at fixed x .

The constants combine as $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \cdot \frac{k^+}{2\pi} \cdot \frac{1}{k^+} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+}$, so both parts become

$$\frac{d^3 N_{a,b}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \int_{\lambda}^{1-\lambda} d\xi \int d^2 r e^{iq \cdot r} \frac{T(r)}{\xi D} \mathcal{U}_{a,b}.$$

3. Part (b): finding all the UV terms

3.1 The regions of the r integral

region	behaviour at fixed $\xi \in (0, 1)$	verdict
$r \rightarrow 0$ ($z \rightarrow x$)	$T \sim \frac{1}{r^2}$ and $\frac{1}{\xi D} \rightarrow \frac{1}{s^2}$; $\int d^2 r \frac{1}{r^2}$ diverges logarithmically	UV divergent
$r \rightarrow s$ ($z \rightarrow y$)	$D \rightarrow M^2 > 0$; the $\frac{1}{(s-r)^2}$ terms in T_2, T_5, T_6 have numerators vanishing linearly, so at worst $\frac{1}{ r-s }$	finite
$ r \rightarrow \infty$	a $\frac{1}{r^2}$ tail, cut off by the oscillation $e^{iq \cdot r}$ at $ r \sim \frac{1}{q}$	finite for $\bar{\xi} > 0$

3.2 The UV part of each block

As $r \rightarrow 0$, only the $O(r^{-2})$ pieces survive the angular average; the $O(r^{-1})$ pieces are odd in $\hat{r}$ and drop out:

block	UV piece
T_1	$d_{\perp} \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4}$
T_2	$\frac{\xi}{\bar{\xi}} \frac{P \cdot s}{P^2 r^2}$
T_3	$-2 \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4}$
T_4	$\frac{\bar{\xi}}{\xi} \frac{P \cdot s}{P^2 r^2}$
T_5	$-2 \xi \frac{(P \cdot r)(s \cdot r)}{P^2 r^4}$
T_6	$+2 \frac{(P \cdot r)(s \cdot r)}{P^2 r^4}$

The coefficients of $\frac{(P \cdot r)(s \cdot r)}{r^4}$ add to $d_{\perp} \xi \bar{\xi} - 2 \bar{\xi} - 2 \xi + 2 = d_{\perp} \xi \bar{\xi}$, because $2 - 2 \xi - 2 \bar{\xi} = 0$: the UV pieces of T_3, T_5, T_6 cancel at every point. So

$$T_{\text{UV}}(r) = d_{\perp} \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2 r^2}, \quad \frac{T(r)}{\xi D} \xrightarrow{r \rightarrow 0} \frac{T_{\text{UV}}(r)}{s^2}.$$

Check. $\rho^2 \langle T / (\xi D) \rangle_{\hat{r}}$ tends to $C_{\text{UV}}(\xi) \mathcal{W}$ to 8 digits: at $\xi = 0.1, 0.3, 0.5, 0.8$, $\rho = 10^{-4}$ gives $-8.96518516, -2.89570207, -2.19230770, -4.29692308$, against $-8.96518519, -2.89570208, -2.19230769, -4.29692308$. After subtracting T_{UV}/s^2 the remainder falls as ρ^2 .

3.3 Subtract and add

$$\frac{T(r)}{\xi D} = \underbrace{\frac{T_{\text{UV}}(r)}{s^2}}_{\text{UV, done in } d=2-2\epsilon} + \underbrace{\left[\frac{T(r)}{\xi D} - \frac{T_{\text{UV}}(r)}{s^2} \right]}_{\text{finite, done at } d_{\perp}=2}.$$

4. Dimensional regularisation of the UV part

4.1 The scalar master

The general formula is $\int d^d r e^{iq \cdot r} (r^2)^{-a} = \pi^{d/2} 2^{d-2a} \frac{\Gamma(d/2-a)}{\Gamma(a)} (q^2)^{a-d/2}$. With $a = 1$ and $d = 2 - 2\epsilon$:

$$\int d^d r \frac{e^{iq \cdot r}}{r^2} = \pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi} \right)^{\epsilon}, \quad \Gamma(-\epsilon) = -\frac{1}{\epsilon} - \gamma_E + O(\epsilon).$$

The integral has mass dimension 2ϵ , so multiply by $\mu^{-2\epsilon}$. Then with $\mu_{\overline{\text{MS}}}^2 = 4\pi\mu^2 e^{-\gamma_E}$:

$$M_0 \equiv \mu^{-2\epsilon} \int d^d r \frac{e^{iq \cdot r}}{r^2} = -\pi \left[\frac{1}{\epsilon} + \gamma_E + \log \frac{q^2}{4\pi\mu^2} \right] = \pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2}{q^2} \right], \quad \frac{1}{\epsilon_{\text{UV}}} \equiv -\frac{1}{\epsilon}.$$

The UV region converges only for $\epsilon < 0$, so $\frac{1}{\epsilon_{\text{UV}}}$ is the positive UV pole. Two checks:

- $\epsilon M_0 \rightarrow -\pi$ exactly.
- With a hard cutoff $|r| > \rho$, the same integral is $-\pi \log \frac{q^2 \rho^2 e^{2\gamma_E}}{4}$. The Bessel quadrature gives 56.9502112 for $\rho = 10^{-4}$, $q = 1.3$, and 93.2910566 for $\rho = 10^{-6}$, $q = 0.4$, both matching that formula.

4.2 The tensor master

In any dimension $\frac{r^i r^m}{r^4} = \frac{1}{2} \left[\frac{\delta^{im}}{r^2} - \partial^i \frac{r^m}{r^2} \right]$. Integrating by parts, $\int e^{iq \cdot r} \partial^i f = -iq^i \int e^{iq \cdot r} f$. Differentiating the scalar master, $\int e^{iq \cdot r} \frac{r^m}{r^2} = -i \partial_{q^m} \left[\pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi} \right)^\epsilon \right] \rightarrow 2\pi i \frac{q^m}{q^2}$. Therefore

$$M_2^{im} \equiv \mu^{-2\epsilon} \int d^d r e^{iq \cdot r} \frac{r^i r^m}{r^4} = \frac{\delta^{im}}{2} M_0 - \pi \hat{q}^i \hat{q}^m.$$

Trace check: $\frac{d}{2} M_0 - \pi = M_0 - \epsilon M_0 - \pi = M_0$. In the cutoff scheme the $\hat{q}\hat{q}$ coefficient comes out -3.14159265359 .

4.3 Contract, keeping $d_\perp$ symbolic

$$\int d^d r e^{iq \cdot r} \frac{T_{\text{UV}}}{s^2} = \frac{1}{P^2 s^2} \left[d_\perp \xi \bar{\xi} \left(\frac{P \cdot s}{2} M_0 - \pi (P \cdot \hat{q})(s \cdot \hat{q}) \right) + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) (P \cdot s) M_0 \right].$$

Now $\frac{d_\perp}{2} M_0 = (1 - \epsilon) M_0 = M_0 + \pi + O(\epsilon)$. This is the one place where setting $d_\perp = 2$ too early would lose a finite term. With $\hat{q} = \hat{k}$:

$$\int d^d r e^{iq \cdot r} \frac{T_{\text{UV}}}{s^2} = \pi \mathcal{W} C_{\text{UV}}(\xi) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2}{\xi^2 k^2} \right] + \pi \xi \bar{\xi} \mathcal{T},$$

$$\mathcal{W} = \frac{P \cdot s}{P^2 s^2}, \quad \mathcal{T} = \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2}.$$

(sympy: the difference from $C_{\text{UV}}(P \cdot s) M_0$ is exactly $\pi \xi \bar{\xi} [P \cdot s - 2(P \cdot \hat{q})(s \cdot \hat{q})]$.)

5. Part (b) after the z integration

Pulling one π into the prefactor turns $\frac{q^4}{16\pi^5}$ into $\frac{q^4}{16\pi^4}$:

$$\frac{d^3 N_b}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \int_\lambda^{1-\lambda} d\xi \left\{ \mathcal{W} C_{\text{UV}}(\xi) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2}{\xi^2 k^2} \right] + \xi \bar{\xi} \mathcal{T} + \frac{1}{\pi} \mathcal{I}_b(\xi) \right\} \mathcal{U}_b,$$

$$\mathcal{I}_b(\xi) = \int d^2 r e^{i\xi k \cdot r} \left[\frac{T(r)}{\xi D} - \frac{T_{\text{UV}}(r)}{s^2} \right]_{d_\perp=2} \quad (\text{finite, no } \epsilon).$$

6. Part (b): the p^+ integration

6.1 The closed-form terms

The two moments of C_{UV} over $\xi \in [\lambda, 1 - \lambda]$, as $\lambda \rightarrow 0$:

term	$\int d\xi$	$\int d\xi \log \frac{1}{\xi^2}$
$\xi \bar{\xi}$	$\frac{1}{6}$	$\frac{5}{18}$
$\frac{\xi}{\xi}$	$L - 1$	$L^2 - 2$
$\frac{\bar{\xi}}{\xi}$	$L - 1$	$\frac{\pi^2}{3} - 2$
C_{UV}	$2L - \frac{11}{6}$	$L^2 + \frac{\pi^2}{3} - \frac{67}{18}$

How each entry is obtained:

- $\int_{\lambda}^{1-\lambda} \frac{\xi}{\xi} d\xi = \int \left(\frac{1}{\xi} - 1 \right) = \log \frac{1-\lambda}{\lambda} - (1 - 2\lambda) \rightarrow L - 1$, and the same for $\frac{\bar{\xi}}{\xi}$.
- For the log moment of $\frac{\xi}{\xi}$, put $v = \bar{\xi}$: $-2 \int_{\lambda}^{\frac{(1-v)\log v}{v}} dv = -2[F(1) - F(\lambda)]$ with $F = \frac{1}{2} \log^2 v - v \log v + v$, which gives $L^2 - 2$.
- For $\frac{\bar{\xi}}{\xi}$: $-2 \left[\int_0^1 \frac{\log(1-\xi)}{\xi} d\xi - \int_0^1 \log(1-\xi) d\xi \right] = -2 \left[-\frac{\pi^2}{6} + 1 \right]$.
- For $\xi \bar{\xi}$: $-2 \int_0^1 v(1-v) \log v dv = -2 \left(-\frac{1}{4} + \frac{1}{9} \right) = \frac{5}{18}$.

Quadrature at $\lambda = 10^{-7}$ gives 30.40285817 and 259.3606566, against 30.40285797 and 259.3606533 from the formulas. Also $\int_0^1 \xi \bar{\xi} d\xi = \frac{1}{6}$ for the $\mathcal{T}$ term. So the closed-form part of (b) is

$$\mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \left\{ \mathcal{W} \left[\left(2L - \frac{11}{6} \right) \left(\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{MS}^2}{k^2} \right) + L^2 + \frac{\pi^2}{3} - \frac{67}{18} \right] + \frac{1}{6} \mathcal{T} \right\} \mathcal{U}_b.$$

6.2 The remainder $\mathcal{I}_b$: can the plus prescription handle it?

The poles in ξ come from T_4 ($\frac{\xi}{\xi}$) and T_2 ($\frac{\xi}{\xi}$). The plus prescription $\int \frac{d\xi}{\xi} [g(\xi) - g(0)]$ is legitimate only if $g(\text{endpoint})$ is finite, so everything depends on how $\mathcal{I}_b$ behaves at each end. The 2D oscillatory quadrature, with s, P, k fixed:

ξ	$\xi \text{ Re } \mathcal{I}_b$	slope in $\log \frac{1}{\xi}$	$\bar{\xi}$	$\bar{\xi} \text{ Re } \mathcal{I}_b$	slope in $\log \frac{1}{\bar{\xi}}$
10^{-2}	2.60199		10^{-2}	15.0395	3.164
$3 \cdot 10^{-3}$	2.61189	0.0082	$3 \cdot 10^{-3}$	18.7812	3.108
10^{-3}	2.61435	0.0022	10^{-3}	22.1549	3.071
$3 \cdot 10^{-4}$	2.61517	0.0007	$3 \cdot 10^{-4}$	25.8392	3.060
10^{-4}	2.61539	0.0002	10^{-4}	29.2387	3.094

In this configuration $-\pi\mathcal{W} = 3.061039$. So:

$$\mathcal{I}_b \xrightarrow{\xi \rightarrow 0} \frac{\beta_0}{\xi}, \quad \mathcal{I}_b \xrightarrow{\xi \rightarrow 1} \frac{-\pi\mathcal{W} \log \frac{1}{\xi} + \beta_1}{\bar{\xi}}.$$

- **At $\xi \rightarrow 0$:** a simple pole with finite $g(0) = \beta_0$. **The plus prescription works.**
- **At $\xi \rightarrow 1$:** $g(\xi) = \bar{\xi} \mathcal{I}_b$ itself diverges like $\log \frac{1}{\xi}$. **One plus distribution is not enough.**

6.3 Where the $\log \frac{1}{\xi}$ comes from

As $\bar{\xi} \rightarrow 0$ the r integral has three scales: $|s|$, the mass $M = |s| \sqrt{\xi/\bar{\xi}} \rightarrow \infty$, and the oscillation length $\frac{1}{q} = \frac{1}{\xi k} \gg M$. At fixed r the integrand is just $\frac{1}{\xi}$ times something finite; the logarithm is generated entirely at large $|r|$:

- in $|s| < |r| < M$, T_2 (whose weight $\frac{\xi}{\xi}$ is large) has a $\frac{1}{r^2}$ tail, giving $\frac{\pi\mathcal{W}}{2\xi} \log \frac{1}{\xi}$;
- in $M < |r| < \frac{1}{q}$, the pieces of T_3, T_4, T_6 that grow like $|r|$ give a second $\frac{1}{r^2}$ tail and another $\frac{\pi\mathcal{W}}{2\xi} \log \frac{1}{\xi}$.

So the true integrand carries $+\frac{\pi\mathcal{W}}{\xi} \log \frac{1}{\xi}$. The UV counterterm $\frac{T_{UV}}{s^2}$ keeps its $\frac{1}{r^2}$ all the way from 0 to $\frac{1}{q}$ and so carries $\frac{2\pi\mathcal{W}}{\xi} \log \frac{1}{\xi}$ — this is the $\log \frac{1}{\xi^2}$ of §5. Their difference, $-\frac{\pi\mathcal{W}}{\xi} \log \frac{1}{\xi}$, is what $\mathcal{I}_b$ carries, as the numbers show.

6.4 The plus prescription, done correctly

You need these two distributions at $\xi \rightarrow 1$ (checked against direct cut-off integration), plus the ordinary one at $\xi \rightarrow 0$:

$$\frac{1}{\xi} = \left[\frac{1}{\xi} \right]_+ + \delta(\xi) L, \quad \frac{\log \frac{1}{\xi}}{\xi} = \left[\frac{\log \frac{1}{\xi}}{\xi} \right]_+ + \frac{1}{2} \delta(\xi) L^2, \quad \frac{1}{\xi} = \left[\frac{1}{\xi} \right]_+ + \delta(\xi) L.$$

Applying them:

$$\int_{\lambda}^{1-\lambda} d\xi \mathcal{I}_b(\xi) = \beta_0 L + \beta_1 L - \frac{\pi \mathcal{W}}{2} L^2 + \mathcal{F}_b + O(\Lambda),$$

$$\mathcal{F}_b = \int_0^1 d\xi \left[\mathcal{I}_b(\xi) - \frac{\beta_0}{\xi} - \frac{\beta_1 - \pi \mathcal{W} \log \frac{1}{\xi}}{\xi} \right], \quad \beta_0 = \lim_{\xi \rightarrow 0} \xi \mathcal{I}_b, \quad \beta_1 = \lim_{\xi \rightarrow 1} \left[\bar{\xi} \mathcal{I}_b + \pi \mathcal{W} \log \frac{1}{\bar{\xi}} \right].$$

The bracket in $\mathcal{F}_b$ is integrable at both ends by construction.

6.5 Part (b): final answer

Add the closed-form part (6.1) to $\frac{1}{\pi} \times (6.4)$. The two L^2 terms combine, $L^2 - \frac{1}{2} L^2 = \frac{1}{2} L^2$:

$$\frac{d^3 N_b}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \left\{ \mathcal{W} \left[\left(2L - \frac{11}{6} \right) \left(\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{MS}}^2}{k^2} \right) + \frac{1}{2} L^2 + \frac{\pi^2}{3} - \frac{67}{18} \right] + \frac{1}{6} \mathcal{T} + \frac{\beta_0 + \beta_1}{\pi} L + \frac{\mathcal{F}_b}{\pi} \right\} \mathcal{U}_b$$

7. Part (a): is there a UV divergence? No.

The colour factor vanishes as $z \rightarrow x$: $\mathcal{U}_a \propto U^{ce}(z) - U^{ce}(x) = r \cdot \partial U^{ce}(x) + O(r^2)$. Two powers are gained against the $\frac{1}{r^2}$ of T_{UV} :

1. one from the linear vanishing, against $d^2 r = |r| d|r| d\phi$;
2. one more because the linear term is odd in $\hat{r}$ while T_{UV} is even, so the angular average kills it.

Check (with a smooth adjoint field U): $\rho^2 \times \langle \text{integrand of (a)} \rangle_{\hat{r}}$ is 4.18×10^{-5} , 10^{-7} , 10^{-9} , 10^{-11} at $\rho = 10^{-2}, \dots, 10^{-5}$ — it falls as ρ^2 . There is nothing to regularise: no counterterm, no $\overline{MS}$, $d_{\perp} = 2$.

8. Part (a): the p^+ integration

Since there is no subtraction, the whole r integral is finite at each ξ :

$$\frac{d^3 N_a}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \frac{1}{\pi} \int_{\lambda}^{1-\lambda} d\xi \mathcal{I}_a(\xi), \quad \mathcal{I}_a(\xi) = \int d^2 r e^{i\bar{\xi} k \cdot r} \frac{T(r)}{\bar{\xi} D} \mathcal{U}_a(x + r).$$

The endpoint structure is the same as in (b):

- $\xi \rightarrow 0$: a simple pole $\frac{\alpha_0}{\xi}$. The only $\frac{1}{\xi}$ is T_4 , whose r integral at $\xi = 0$ converges at $r \rightarrow 0$ (§7), at $r \rightarrow s$ (T_4 vanishes there) and at large r (it oscillates as $e^{ik \cdot r}$). The plus prescription works.
- $\xi \rightarrow 1$: the logarithm is generated at large $|r|$, exactly as in §6.3, but now with no counterterm. The whole $+\frac{\pi \mathcal{W}}{\xi} \log \frac{1}{\xi}$ therefore survives, multiplied by the colour factor at large $|r|$:

$$\mathcal{I}_a \xrightarrow{\xi \rightarrow 1} \frac{\pi \mathcal{W} \mathcal{U}_a^{\infty} \log \frac{1}{\xi} + \alpha_1}{\bar{\xi}}, \quad \mathcal{U}_a^{\infty} = [U^{ab'}(x') - U^{ab'}(w')] (-f^{adc} f^{fbc} U^{db}(x) + N_c U^{af}(x)) \rho^{b'}(x') \rho^f(y).$$

This assumes $U(z) \rightarrow 1$ outside the target (the usual finite-target convention). In the test configuration the colour matrix inside $\mathcal{U}_a^{\infty}$ has norm 1.30, so it does not vanish. Two distributions are needed, as in §6.4.

$$\frac{d^3 N_a}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \left\{ \frac{1}{2} \mathcal{W} \mathcal{U}_a^{\infty} L^2 + \frac{\alpha_0 + \alpha_1}{\pi} L + \frac{\mathcal{F}_a}{\pi} \right\}$$

$$\mathcal{F}_a = \int_0^1 d\xi \left[\mathcal{I}_a - \frac{\alpha_0}{\xi} - \frac{\alpha_1 + \pi \mathcal{W} \mathcal{U}_a^{\infty} \log \frac{1}{\xi}}{\bar{\xi}} \right], \quad \alpha_0 = \lim_{\xi \rightarrow 0} \xi \mathcal{I}_a, \quad \alpha_1 = \lim_{\xi \rightarrow 1} \left[\bar{\xi} \mathcal{I}_a - \pi \mathcal{W} \mathcal{U}_a^{\infty} \log \frac{1}{\bar{\xi}} \right].$$

(Here $\alpha_{0,1}$ and $\mathcal{F}_a$ already contain the colour factor, since it sits inside the r integral.)

9. One caveat for the later x integral

The $\delta(\xi)$ terms, $\beta_0 L$ in (b) and $\alpha_0 L$ in (a), are finite at fixed x, y . But when you later do $\int_x$ they are **logarithmically divergent at large $|x|$** . Near $r \approx s$ with $|s|$ large, T_4 's residue is $\frac{P \cdot (s-r)}{2P^2 s^2}$, and $\int d^2 u e^{ik \cdot u} \frac{-P \cdot u}{u^2} = -2\pi i \frac{P \cdot k}{k^2}$. So

$$\beta_0 \xrightarrow{|s| \rightarrow \infty} -\frac{i\pi (P \cdot k)}{P^2 k^2} \frac{e^{ik \cdot s}}{s^2},$$

and together with $e^{-ik \cdot (w' - x)}$ the total phase is $e^{-ik \cdot w'} e^{ik \cdot y}$, with no x in it. So $\int d^2 x \beta_0 \sim \int \frac{d^2 x}{(y-x)^2}$ diverges. Before the r integral, this is a $\frac{1}{x^2}$ tail measured at -0.0457352 against a predicted -0.0457352 . Physically it is the $p^+ \rightarrow 0$ gluon far from everything else, and its colour factor there ($N_c[U(y) - 1]$ in (b)) is not the LO one. None of the formulas above is affected, because the x integral is not done here. When you do it, this piece has to be either combined with the other Group I rows, or regulated by keeping ξ nonzero through the x integral.

10. Summary

	UV at $z \rightarrow x$	p^+ at $\xi \rightarrow 0$	p^+ at $\xi \rightarrow 1$	double log
(b)	one pole, $(2L - \frac{11}{6}) \frac{1}{\epsilon_{UV}}$	simple pole: $[\frac{1}{\xi}]_+$	$\frac{\log}{\xi}$: two distributions	$\frac{1}{2} \mathcal{W} L^2$
(a)	none	simple pole: $[\frac{1}{\xi}]_+$	$\frac{\log}{\xi}$: two distributions	$\frac{1}{2} \mathcal{W} \mathcal{U}_a^\infty L^2$

Checks: `src/rowV_final.py` (steps 1–8: the six blocks against the `.tex`, denominator/phase/prefactor in `sympy`, the UV residue, both masters incl. the cutoff cross-check, the $d_\perp$ -symbolic contraction, the ξ moments, the part-(a) UV scan, the β_0 tail), output in `RESULTS_rowV_final.txt`; `src/row28_rem.py` (the endpoint scans of $\mathcal{I}_b$), output in `RESULTS_rowV_remainder_rerun.txt`; `src/rowB2_2a_full.py` (the distribution identities); `src/rowV_softx.py` (the $\frac{1}{x^2}$ tail).